Fusion of the completed Higman–Sims families

A certificate using the acquired full automorphism action

Let G be the retained permutation group on 100 points. The accepted normalizer certificate supplies $G \trianglelefteq N$, with $|G| = 44352000$, $|N| = 88704000$ and $C_N(G) = 1$. The acquired complete order $|\operatorname{Aut}(G)| = 88704000$ therefore identifies the conjugation action of N with the full automorphism group. This is an explicit acquired premise; the present calculation does not repeat an automorphism-group or subgroup census.

Proposition 1. *The simultaneous transporter in* `fusion_conjugation.g` *decides equivalence of ordered subgroup families under conjugation by N and arbitrary permutations of their types. Every positive answer contains a literal permutation of the 100 points and a type permutation.*

Proof. Fix a type permutation, so that the desired target family is $(Q_1, \ldots, Q_r)$. Begin with the original family $(P_1, \ldots, P_r)$, accumulated conjugator $a = 1$, and acting group $C = N$. At step i, find $x \in C$ carrying the current ith subgroup to Q_i; if none exists, this type permutation is impossible. Conjugate every current subgroup by x, replace a by ax, and replace C by $N_C(Q_i)$. The earlier target subgroups remain fixed. Inductively, every possible completion differs from this chosen partial transporter by an element of the new C: if two elements carry the same subgroup to Q_i, their quotient on the target side normalizes Q_i. Thus a failed step excludes every completion for this type permutation, and success carries the entire family. Trying every type permutation is exhaustive. Orders of individual groups, pair intersections and the full intersection are used only as necessary rejection tests. The code checks $a \in N$ and $P_i^a = Q_i$ literally before returning. □

The completed anchor inventory covers every one of the 589 acquired subgroup classes exactly once: 529 by completed enumeration and 60 by the separately proved proper-generation bound. The latter have no full-G families and are recorded separately, not represented by invented empty enumerations. For each retained full family of panels $(T_1, \ldots, T_r)$ with Borel B, form $P_i = \langle T_j : j \neq i \rangle$. The acquired intersection property gives $\bigcap_i P_i = B$ and $T_i = \bigcap_{j \neq i} P_j$; the calculation also checks these equalities on each actual family.

Successive representatives are compared by the proposition. A failed comparison separates two orbits; a positive witness assigns the new family to an existing orbit. This partitions precisely the completed input. The output concerns geometries together with their specified G action, modulo $\operatorname{Aut}(G)$ and types. Identification with the source's bare incidence classes requires its separate isomorphism comparison and is not inferred from this partition alone.

The transporter is the earlier simultaneous-family method written in the already acquired permutation action, with its actual conjugator retained. Small controls compare it with exhaustive enumeration of all acting permutations and all type permutations, including the outer action of S_4 on A_4. The complete target and every input receipt remain separately bound. No manuscript, conjecture or Lean credit is assigned by this method note.